Nuclear Stability

The band of stability and predicting decay

Name _____ Class _____ Date _____

RULES AND DATA

$N = A - Z$ · above the band (too many n): β^- · below (too many p): β^+ or electron capture · Z above 83: α

Stable isotopes: Co-59, F-19, P-31, H-1 and H-2 · magic numbers 2, 8, 20, 28, 50, 82, 126

Atomic numbers: H 1 · He 2 · O 8 · F 9 · P 15 · Ca 20 · Fe 26 · Co 27 · Rn 86 · Po 84 · Th 90 · Pb 82 · U 92

Balance mass numbers and atomic numbers in every nuclear equation.

A. Ideas

1. (a) What is the strong nuclear force? (b) Explain why heavy nuclei need more neutrons than protons to be stable. (c) Why are all nuclei with Z above 83 radioactive?

2. Calculate N/Z for: (a) helium-4 (b) oxygen-16 (c) iron-56 (d) lead-206. Describe the trend.

B. Predicting decay

3. Predict the type of decay and write the nuclear equation for: (a) cobalt-60 (b) fluorine-18 (c) radon-222 (d) phosphorus-32

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4. Calcium-40 and lead-208 are both described as “doubly magic”. Explain what this means and why it matters.

C. Decay series

5. Thorium-232 decays through a series to lead-208. How many α and β^- decays occur? Show your reasoning.

6. Uranium-235 decays through a series to lead-207. How many α and β^- decays occur?

D. Applying the ideas

7. Fluorine-18 is used in PET scans. Explain, using its N/Z ratio, why it emits positrons.

E. Challenge

8. Tritium (hydrogen-3) is radioactive, but hydrogen-1 and hydrogen-2 are stable. Calculate the N/Z ratio of tritium, predict its decay, and write the equation.

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